

8-Phase N-D-C Structural Analysis: Lehman Brothers (Q2 2008)

This analysis maps Lehman Brothers' institutional functions onto 8 structural phases using the N-D-C framework. Each phase is evaluated by its balance between Definition (D: constraints, limits, risk controls in force) and Contribution (C: outputs, deployments, risk exposures produced). The chain average balance is the system-level diagnostic.

The critical finding: Phases 1 and 7 (capital adequacy and regulatory compliance) appear Coherent, matching the regulatory picture. Six operational phases are in Breakdown or at the critical threshold. Phase 3 entered Breakdown in Q1 2007 and remained there for 8 consecutive quarters before failure. Phase 0 (Funding Base) was structurally in Breakdown throughout 2007-2008.

49.9%

Chain average balance (π axis)
6 phases below 61.8% coherence floor

Primary bottleneck: Phase 3: Leverage and Risk Management (17.2%). Lehman's stated leverage limit of 25x was routinely overridden. Gross leverage reached 30.7x in Q1 2008. D (enforced risk constraints) was structurally absent despite formal limit frameworks existing on paper. Risk committee waivers were granted routinely rather than positions reduced.

Phase Balance Scores: Q2 2008 Snapshot (Final Full Quarter Before Failure)

#	PHASE	BALANCE	STATUS	D / C	KEY STRUCTURAL SIGNAL
0	Funding Base and Liability Structure Diversity and maturity profile of the liability base	28.3%	BREAKDOWN	D 85 / C 300	~\$188B in overnight repo requiring daily rollover. No effective maturity diversification. C (funding volume) vastly exceeded D (stability constraints). Any counterparty withdrawal was immediately terminal.
1	Capital Adequacy (Reported) Δ Regulatory capital against risk-weighted assets:	97.4%	DECEPTIVE SIGNAL	D 195 / C 190	Tier 1 ~11% appeared adequate. Hidden: Repo 105 removed \$50.4B from balance sheet at quarter-end. True gross leverage 30.7x vs reported 12.1x. Valukas

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	Repo 105 manipulation active				Report called this actionable balance sheet manipulation.
2	Asset Quality and Portfolio Concentration Capital deployed into loans, CRE, mortgages, and leveraged positions	25.9%	BREAKDOWN	D 75 / C 290	\$32.6B commercial real estate + \$24.8B residential mortgage exposure. No effective concentration limits. Originate-to-hold replaced originate-to-distribute. \$8.2B write-downs in Q2 2008 alone.
3	Leverage and Risk Management Gross leverage and risk exposure managed through limits and stress testing	17.2%	BREAKDOWN: PRIMARY	D 55 / C 320	Gross leverage 30.7x vs 25x stated target. Risk committee waivers granted routinely. VaR limits exceeded without position reduction. D (enforced constraints) structurally absent. Model detected Breakdown in Q1 2007.
4	Liquidity Pool and Funding Access Short-term obligations met through liquid asset pool	43.2%	BREAKDOWN	D 185 / C 80	Claimed \$45.5B liquidity pool. Valukas found substantial portion encumbered. No FDIC insurance; limited Fed emergency access for broker-dealers. Daily repo rollover of \$188B dwarfed usable liquidity.
5	Earnings and Net Revenue Asset portfolio and fee income converted into net revenue	61.3%	CRITICAL THRESHOLD	D 155 / C 95	Record \$4.2B earnings in 2007 masked structural deterioration. Q2 2008: first quarterly loss (-\$2.8B). Q3 2008: -\$3.9B loss announced 5 days before failure. Revenue model required continuous asset price appreciation.
6	Market Confidence and Counterparty Trust Institutional credibility and repo counterparty relationships	30.6%	BREAKDOWN	D 180 / C 55	Korea Development Bank deal collapsed September 9. Barclays and BofA declined without government backstop. Repo counterparties refused to roll agreements. A rated until 6 days before failure.
7	Regulatory Oversight and Reporting ⚠️	94.9%	DECEPTIVE SIGNAL	D 195 / C 185	SEC CSE program allowed self-monitoring with internal models. Repo 105 not

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	SEC CSE program compliance and supervisory obligations				identified by examiners. No statutory resolution authority for broker-dealers. FCIC concluded regulatory oversight fundamentally inadequate.

Technical Dashboard

Intervention Analysis

Resolution Analysis

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